MODULE 2 · LESSON 1 · BOOK PP. 4–14

Quiz · Organisms and Their Relationships

20 marks · 25 minutes الكائنات الحية وعلاقاتها

Name _____ **Class** _____ **Score** / 20

SECTION A · MULTIPLE CHOICE · 6 × 2 MARKS

1 Why do ecologists use models?ليه علماء البيئة بيستخدموا النماذج؟ **2 marks**

- **A** Field studies have too many variables; models let them control variables and add them one at a time
- **B** Models remove the need for any observation
- **C** Models are always more accurate than real data
- **D** Models are cheaper than paper

2 Which situation is the best example of a limiting factor?أفضل مثال لعامل محدد؟ **2 marks**

- **A** A deer drinks water every day.
- **B** Less rainfall reduces the size of a grass population.
- **C** Two birds use different parts of a tree.
- **D** One species benefits while another is unaffected.

3 What is the difference between a biological community and an ecosystem?الفرق بين المجتمع والنظام البيئي؟ **2 marks**

- **A** They are the same
- **B** An ecosystem has only one species
- **C** An ecosystem includes the community AND the abiotic factors
- **D** A community includes abiotic factors but an ecosystem does not

4 A bird feeds on insects living on a large mammal. The bird gains food; the mammal is neither helped nor harmed. Which interaction is this?طائر يأكل حشرات على حيوان كبير... **2 marks**

- **A** Parasitism
- **B** Competition
- **C** Mutualism
- **D** Commensalism

5 A photo shows moss and mushrooms growing on a fallen log in a forest. Which statement about the photo is TRUE?صورة طحالب وعيش غراب على جذع في غابة. أي عبارة صحيحة؟ **2 marks**

- **A** It shows a place outside the biosphere.
- **B** It shows only abiotic factors.
- **C** It shows organisms that do not depend on their environment.
- **D** It shows part of the biosphere, where organisms depend on living and nonliving factors.

6 A swamp plant is moved to a dry desert and dies. What is the best explanation?نبات مستنقعات اتنقل للصحراء ومات. التفسير؟ **2 marks**

- **A** It could not adjust to the new abiotic factors
- **B** The desert had too much rain
- **C** It had too many predators
- **D** It was a biotic factor

SECTION B · WRITTEN ANSWERS · 2 × 4 MARKS

7 Explain the difference between a population and a biological community. Give an example of each.الفرق بين الجماعة والمجتمع مع مثال. **4 marks**

8 Explain why the biosphere is described as a thin layer and not the whole Earth.فسري لماذا المحيط الحيوي طبقة رقيقة. **4 marks**

Total: ____ / 20

MODULE 2 · LESSON 2 · BOOK PP. 15–18

Quiz · Flow of Energy in an Ecosystem

20 marks · 25 minutes تدفق الطاقة في النظام البيئي

Name _____ **Class** _____ **Score** / 20

SECTION A · MULTIPLE CHOICE · 6 × 2 MARKS

1 A fungus grows on a dead log. Which statement best describes its source of energy?فطر على جذع ميت... **2 marks**

A It produces its own food from inorganic substances.
B It changes soil nutrients into sunlight.
C It gets energy from organic matter in the dead organism.
D It gets energy directly from sunlight.

2 Why is only about 10% of energy passed to the next level?ليه 10% بس؟ **2 marks**

A Most energy is used for cellular processes or released as heat
B Decomposers steal 90%
C Organisms refuse to eat
D Energy is recycled to producers

3 A mockingbird eats seeds, fruits and insects. Why is it a heterotroph?الطائر المحاكي... **2 marks**

A It makes food from sunlight.
B It gets energy by eating other organisms.
C It releases nutrients from dead organisms.
D It makes organic molecules from inorganic substances.

4 A food web shows: acacia → gazelle, gazelle → cheetah, gazelle → hyena, grasshopper → bird. Which is one complete energy pathway from a producer?شبكة غذائية... **2 marks**

A grasshopper → acacia → bird
B hyena → gazelle → acacia
C cheetah → gazelle → acacia
D acacia → gazelle → cheetah

5 Why do autotrophs make up the first trophic level?لماذا المنتجات في المستوى الأول؟ **2 marks**

A They produce food that makes energy available to other organisms.
B They get energy from all consumers.
C They eat organisms at higher levels.
D They send energy back to the Sun.

6 Producers capture 10,000 J. About how much energy reaches the secondary consumers?المنتجين 10000 جول، كم يصل للمستهلك الثانوي؟ **2 marks**

A 10 J
B 1,000 J
C 9,000 J
D 100 J

SECTION B · WRITTEN ANSWERS · 2 × 4 MARKS

7 Predict what would happen to a forest if all decomposers disappeared.توقعي لو اختفت المحللات. **4 marks**

8 Explain the pathway of energy transfer through a pyramid of energy.اشرحي انتقال الطاقة في الهرم. **4 marks**

Total: ____ / 20

MODULE 2 · LESSON 3 · BOOK PP. 19–24

Quiz · Cycling of Matter

20 marks · 25 minutes دورة المادة

Name _____ **Class** _____ **Score** / 20

SECTION A · MULTIPLE CHOICE · 6 × 2 MARKS

1 About 10% of water vapor evaporates from the surfaces of plants. This process is... 10% تتبخر من النباتات... — **2 marks**

A transpiration
B condensation
C runoff
D percolation

2 Which comparison correctly describes photosynthesis and cellular respiration in the carbon–oxygen cycle? قارني البناء الضوئي والتنفس. — **2 marks**

A Both remove CO_2 from the air.
B Photosynthesis returns CO_2; respiration removes it.
C Photosynthesis removes CO_2 from the air; cellular respiration returns CO_2.
D Both release oxygen as their main effect.

3 Which sequence follows nitrogen from the atmosphere until it becomes part of a plant? تسلسل النيتروجين من الجو للنبات... — **2 marks**

A Assimilation → nitrification → nitrogen fixation
B Nitrogen fixation → nitrification → assimilation
C Nitrification → nitrogen fixation → assimilation
D Denitrification → assimilation → nitrogen fixation

4 In the short-term phosphorus cycle, phosphates in solution are cycled... في الدورة القصيرة... — **2 marks**

A from the atmosphere to the soil
B only within the ocean floor
C from rocks directly to animals
D from the soil to producers and then to consumers

5 On a hot day a lake gets smaller and leaves of nearby trees lose water. Which two processes are happening? يوم حار... — **2 marks**

A Condensation and precipitation
B Evaporation and transpiration
C Precipitation and transpiration
D Percolation and runoff

6 Which process moves carbon from the abiotic part (atmosphere) to the biotic part of the ecosystem? أي عملية تنقل الكربون من الجو للكائنات؟ — **2 marks**

A Photosynthesis
B Respiration
C Combustion
D Decay

SECTION B · WRITTEN ANSWERS · 2 × 4 MARKS

7 Explain how burning fossil fuels changes the amount of carbon in the atmosphere. كيف يغير حرق الوقود الأحفوري كمية الكربون في الجو؟ — **4 marks**

8 Describe how a water molecule in the ocean can end up as groundwater. صفي رحلة جزيء ماء من المحيط للمياه الجوفية. — **4 marks**

Total: ____ / 20

MODULE 5 · LESSON 1 · BOOK PP. 30–36

Quiz · Biodiversity

20 marks · 25 minutes التنوع الحيوي

Name _____ **Class** _____ **Score** / 20

SECTION A · MULTIPLE CHOICE · 6 × 2 MARKS

1 Asian ladybird beetles of the same species with different colors show...خنافس الدعسوقة بألوان مختلفة... — **2 marks**

A ecosystem diversity
B genetic diversity
C extinction
D species diversity

2 Which is the BEST example of a direct economic benefit of biodiversity?أفضل مثال لقيمة اقتصادية مباشرة؟ — **2 marks**

A Wood from trees used to build houses
B A grassland preventing soil erosion
C A forest producing oxygen
D A wetland filtering water

3 As you move from the polar regions toward the equator, species diversity...من القطبين للاستواء... — **2 marks**

A stays the same
B disappears
C increases
D decreases

4 Mangrove forests along the UAE coast protect the shore from waves and storms. This is an example of...أشجار القرم تحمي الساحل... — **2 marks**

A overexploitation
B direct economic value
C background extinction
D indirect economic value

5 Area A has 40 bird species; area B has 8 bird species. Which statement is correct?منطقة A فيها 40 نوع... — **2 marks**

A Area A has higher species diversity.
B Area B has higher species diversity.
C Area A has lower biodiversity.
D Both have the same genetic diversity.

6 Fishermen sell fish caught near a coral reef. This is an example of...صيادون يبيعون السمك... — **2 marks**

A biological magnification
B indirect economic value
C aesthetic value only
D direct economic value

SECTION B · WRITTEN ANSWERS · 2 × 4 MARKS

7 Name the three types of biodiversity and give an example of each.اذكري الأنواع الثلاثة مع أمثلة. — **4 marks**

8 Explain why scientists want to protect wild relatives of crop plants.لماذا نحمي الأقارب البرية للمحاصيل؟ — **4 marks**

Total: ____ / 20

MODULE 5 · LESSON 2 · BOOK PP. 37–44

Quiz · Threats to Biodiversity

20 marks · 25 minutes التهديدات للتنوع الحيوي

Name _____ **Class** _____ **Score** / 20

SECTION A · MULTIPLE CHOICE · 6 × 2 MARKS

1 Which kind of extinction concerns scientists today؟أي انقراض يقلق العلماء — **2 marks**

A Extinction of dinosaurs
B The recent increase in the rate of extinction
C Extinction caused only by volcanoes
D Background extinction

2 More than half of all species on Earth live in...أكثر من نصف الأنواع تعيش في... — **2 marks**

A polar regions
B the open ocean
C tropical rain forests
D deserts

3 Scientists predict climate change will threaten about what percent of all land species by 2050؟نسبة الأنواع المهددة بحلول 2050 — **2 marks**

A 25%
B 5%
C 50%
D 90%

4 A rat species arrives by ship on an island where birds nest on the ground. Why might the birds become extinct؟فئران وصلت لجزيرة... — **2 marks**

A The birds have large populations.
B The rats bring more food for the birds.
C Islands have too many predators already.
D They evolved without predators and cannot escape the rats.

5 A new highway divides a large forest into two smaller parts. Which problem is most likely؟طريق يقسم الغابة... — **2 marks**

A Habitat fragmentation with more edge effects
B Bioremediation
C Biological magnification
D Eutrophication

6 How does an increase in global temperatures threaten biodiversity؟إزاي الاحترار يهدد التنوع — **2 marks**

A Higher temperatures stop extinction
B Warmer temperatures always increase biodiversity
C Only plants are affected
D Species evolved for certain temperature ranges; members that cannot adapt die, affecting species that depend on them

SECTION B · WRITTEN ANSWERS · 2 × 4 MARKS

7 Explain why top predators such as eagles have the highest concentration of DDT.ليه المفترس الأعلى عنده أعلى تركيز DDT؟ — **4 marks**

8 Explain how habitat fragmentation reduces biodiversity.اشرحي كيف تقلل التجزئة التنوع. — **4 marks**

Total: ____ / 20

MODULE 5 · LESSON 3 · BOOK PP. 45–52

Quiz · Conserving Biodiversity

20 marks · 25 minutes الحفاظ على التنوع الحيوي

Name _____ **Class** _____ **Score** / 20

SECTION A · MULTIPLE CHOICE · 6 × 2 MARKS

1 A country builds solar farms instead of new oil power plants. Why is this a better long-term choice?طاقة شمسية بدل البترول... **2 marks**

- **A** Solar energy is a fossil fuel.
- **B** Solar energy is renewable; oil is nonrenewable.
- **C** Both are nonrenewable.
- **D** Oil is replaced faster than it is used.

2 A region has 2000 endemic plant species and has lost 80% of its original habitat. It is a...منطقة بها 2000 نوع متوطن وفقدت 80%... **2 marks**

- **A** buffer zone
- **B** megareserve
- **C** biodiversity hotspot
- **D** corridor

3 In general, the larger the affected area...كل ما المساحة أكبر... **2 marks**

- **A** the longer it takes to recover
- **B** it never recovers
- **C** the faster it recovers
- **D** the recovery time does not change

4 Planting a new tree each time one is cut is an example of...زرع شجرة مكان كل شجرة... **2 marks**

- **A** overexploitation
- **B** biomagnification
- **C** fragmentation
- **D** sustainable use

5 About what percent of the world's land is currently set aside as some type of reserve?نسبة الأرض المحمية؟ **2 marks**

- **A** 90%
- **B** 10–15%
- **C** 50%
- **D** 1–2%

6 Scientists sped up the cleanup by microorganisms by...سرعوا التنظيف بـ... **2 marks**

- **A** adding nutrients to the soil
- **B** removing all oxygen
- **C** adding more fuel
- **D** adding pesticides

SECTION B · WRITTEN ANSWERS · 2 × 4 MARKS

7 Explain why a forest can be renewable in one situation and nonrenewable in another.متى تكون الغابة متجددة ومتى لا؟ **4 marks**

8 Explain how a megareserve with buffer zones helps both biodiversity and local people.كيف تساعد المحمية الكبرى؟ **4 marks**

Total: ____ / 20

MODULE 1 · LESSON 1 · BOOK PP. 58–65

Quiz · The Methods of Science

20 marks · 25 minutes طرق العلم

Name _____ **Class** _____ **Score** / 20

SECTION A · MULTIPLE CHOICE · 6 × 2 MARKS

1 Why are scientific explanations sometimes modified?ليه التفسيرات العلمية بتتغير؟ — **2 marks**

A New information or technology shows earlier explanations were incomplete
B Old explanations are always wrong
C Scientists change their minds randomly
D Laws turn into theories

2 A student tests how the amount of salt affects the boiling time of water. What is the dependent variable?تجربة الملح ووقت الغليان... — **2 marks**

A The type of pot
B The time it takes for the water to boil
C The volume of water
D The amount of salt

3 Which statement about laws and theories is correct?عبارة صحيحة؟ — **2 marks**

A Laws explain why things happen
B Theories become laws with enough evidence
C A theory can explain a law, but theories do not become laws
D Laws and theories are the same

4 A scientist studies how lightning forms in clouds. Which branch is this mainly?عالم يدرس البرق... — **2 marks**

A Physical science only
B Social science
C Life science
D Earth science

5 In the ISS microgravity experiment on plants, what is the independent variable?المتغير المستقل في تجربة الجاذبية الصغرى؟ — **2 marks**

A Microgravity
B Amount of water
C Type of soil
D Plant growth

6 Pilots practicing landings without leaving the ground use a...الطيارين بيتدربوا على... — **2 marks**

A physical law
B hypothesis
C computer simulation
D blind experiment

SECTION B · WRITTEN ANSWERS · 2 × 4 MARKS

7 A student tests whether light color affects how fast pond plants make bubbles. Identify the independent variable, dependent variable, two constants and a control.حددي المتغيرات. — **4 marks**

8 Give two reasons scientists use models, with an example.سببين لاستخدام النماذج مع مثال. — **4 marks**

Total: ____ / 20